

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		neither model has any neutrons ☐ Thomson had electron shells in his model ☐ Thomson did not think atoms are mostly empty space ☒			1	1		
		(ii)		electrons are in shells outside the nucleus ☐ atoms have equal numbers of protons and electrons ☐ atoms have equal numbers of protons and neutrons ☒			1	1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		electrons are outside the nucleus☒ electrons are inside the nucleus☐ there are more protons than electrons in an atom☐ protons are in a nucleus in the centre of an atom☒ atoms have particles with no charge☒ award (2) for three correct award (1) for two correct (and one incorrect) deduct (1) for any additional incorrect answer			2	2		
	(b)			13 protons (1) 14 neutrons (1)		2		2		
				Question 5 total	0	2	4	6	0	0